

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA0607

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO5: Introduction – NumPy Arrays – NumPy Basics- Math – Random – Indexing – Filtering – Statistics – Aggregation – saving data – Data analysis with Pandas – DataFrame – Combining – Indexing – File I/O – Grouping – Filtering – Sorting – Plotting (BL 3)

Assessment Tool Weightage: 10%

QUESTIONS: 03

Total Marks:25

Annexure A

MOCK INTERVIEW QUESTIONS

Code of Conduct:

I VIMALA K 192424276 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	30%	20%	25%	15%	10%	
Q1						
Q2						

Total Marks :

Annexure A

1. You are given a small dataset containing student records with missing marks and inconsistent grade formats. How would you use **NumPy and Pandas** to clean the data, fill missing values, and standardize grades? Explain your approach in simple steps.
2. A company provides you with sales data, and you are asked to find the total and average sales for each product category. How would you use **Pandas grouping and aggregation functions** to solve this problem? Explain the steps clearly.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	25	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Student Data Cleaning and Grade Standardization Using NumPy and Pandas

I would use Pandas and NumPy to clean the student dataset in a few simple steps.

1. Load the dataset using Pandas `read_csv()`.
2. Check for missing marks using `isnull().sum()`.
3. Convert marks into numeric format using `pd.to_numeric()`. Invalid values can be converted into NaN.
4. Fill missing marks using the median of the available marks. The median is useful because it is less affected by extreme values.
5. Standardize grades by removing extra spaces using `strip()` and converting all grades to uppercase using `upper()`. For example, a, A, and a+ can be converted into standard formats such as A and A+.
6. Use NumPy for numerical operations such as calculating the median and checking valid mark ranges.
7. Validate the cleaned data by checking that no marks are missing and that all grades follow the same format.

Example:

```
import pandas as pd
import numpy as np
df = pd.read_csv("students.csv")
df["Marks"] = pd.to_numeric(df["Marks"], errors="coerce")
df["Marks"] = df["Marks"].fillna(df["Marks"].median())
df["Grade"] = df["Grade"].str.strip().str.upper()
print(df)
```

Conclusion:

Thus, Pandas is mainly used for data loading, cleaning, and standardization, while NumPy is used for numerical calculations. This produces a clean and consistent student dataset suitable for further analysis.

2. Sales Analysis Using Pandas Grouping and Aggregation

I would use Pandas grouping and aggregation functions to find the total and average sales for each product category in a few simple steps.

1. Load the dataset using Pandas `read_csv()`.
2. Check the sales data using `head()` and `isnull().sum()` to understand the dataset and identify missing values.
3. Group the data by product category using the `groupby()` function. This separates the sales records into categories such as Electronics, Clothing, and Furniture.
4. Calculate total sales using the `sum()` aggregation function for each category.
5. Calculate average sales using the `mean()` aggregation function for each category.
6. Use the `agg()` function to calculate both total and average sales together.
7. Validate the results by comparing the sum of category-wise sales with the total sales of the original dataset.

Example:

```
import pandas as pd
df = pd.read_csv("sales.csv")
result =
    df.groupby("Product_Category")["Sales"].agg( Total_Sales="sum",
    Average_Sales="mean"
)
print(result)
```

Final Answer:

Thus, Pandas `groupby()` is used to organize sales according to product categories, while `sum()` and `mean()` are used to calculate the total and average sales. The `agg()` function allows both calculations to be performed together, producing a clear summarized sales report for further analysis.